CM52059 : Understanding Deep learning

Coursework

Age and Gender Prediction Using Deep Learning (Model A: Custom CNN & Model B: MobileNetV2 Transfer Learning)

Shared Links for 2 models :

Model A :
https://drive.google.com/file/d/1350HvNBrXZHvxLpicCxS3zY6RZJODJxr/view?usp=drive_link

Model B :
https://drive.google.com/file/d/1tkJLqUISRqQBr5sLbjr9h7Jf15bQ9Lyd/view?usp=drive_link

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1. Introduction

This coursework implements two convolutional neural network (CNN) models for simultaneous age estimation and gender classification from face images. The models are trained on the UTKFace dataset and evaluated on unseen test data. The document demonstrates both custom CNN design from scratch (Model A) and transfer learning approaches (Model B).

2. Data Preprocessing

Dataset

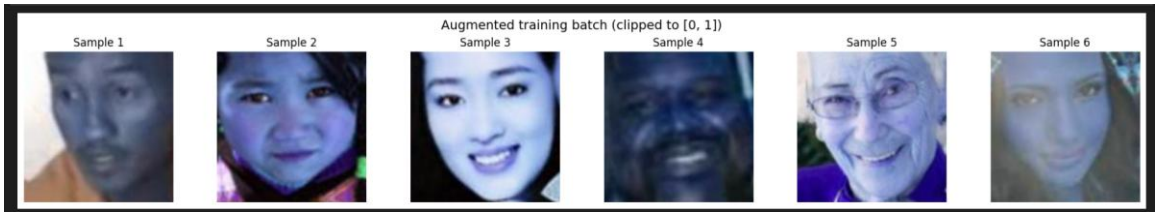
The experiments use 5,000 labelled face images from the UTKFace dataset, each resized to 128×128 RGB resolution. The data is split into 4,000 images for training (80%) and 1,000 images for validation (20%), with an additional held-out set of 2,000 unseen images used only for final testing. Filenames follow the UTKFace convention age_gender_race_date.jpg, from which age and gender labels are parsed.

Data preprocessing

To inspect the raw data, 24 sample images are displayed in a 4×6 grid, each annotated with its age and gender label. All images are then converted to floating-point tensors and normalised to the [0,1] range by dividing pixel intensities by 255.0, while age values are kept in their original scale because mean absolute error (MAE) is reported in years.

Data augmentation

During training, light data augmentation is applied to reduce overfitting and improve generalisation. Each image was horizontally flipped with a probability of 0.5, and a small random rotation of up to 0.03 radians (approximately 1.7 degrees) was applied. This light augmentation added useful variation to the data without damaging fine details, which were important for accurate age prediction. [1]



Data pipeline

The preprocessed and augmented images are loaded using TensorFlow's tf.data API, which enables efficient shuffling, batching, and prefetching during training. Data is fed to the network in mini-batches of 32 images, providing a balance between stable gradient estimates and GPU memory usage while maintaining good training throughput. [2]

3. Model A — Custom CNN

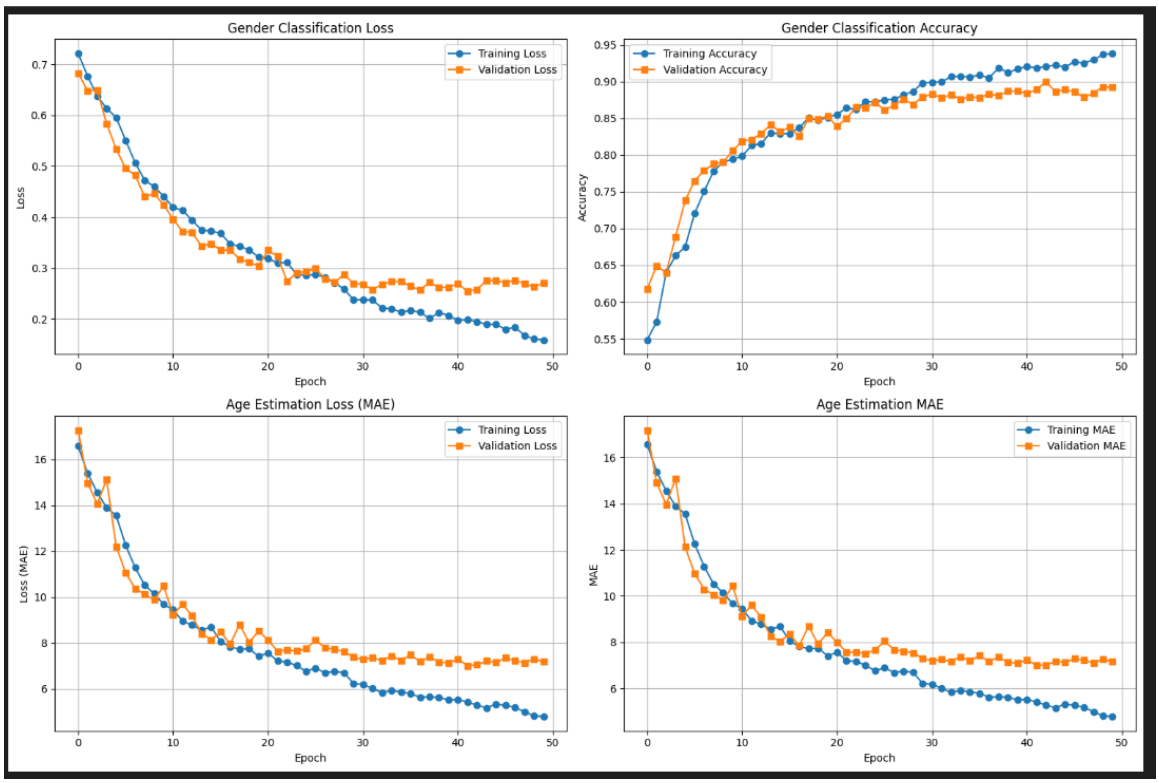
Model A Architecture

This Model A is a simple CNN built from standard components taught in class, following the stacking small convolutions instead of using large kernels.[3] The network uses four convolution–max pooling blocks with 3×3 filters (32, 64, 128, 256 channels), ReLU activations, and 2×2 max pooling to gradually reduce the spatial resolution from 128×128 down to 8×8 while increasing the number of feature maps. This "deep with small filters" strategy is more parameter efficient than a single large 7×7 convolution, reduces overfitting risk, and adds more nonlinear layers so the model can learn richer hierarchical features. . After flattening, the shared features were passed into two dense branches (256 units + 0.3 dropout): a gender head with a sigmoid output and an age head with a ReLU output. This kept the backbone shared while giving each task its own final layer.[4]

Training , Regularisation, and Overfitting Control

Model A was trained with the Adam optimiser (learning rate 0.001) using Keras' builtin compile and fit APIs for multioutput models .[5]The multitask setup uses binary crossentropy with accuracy for gender classification and mean absolute error (MAE) for age regression so that age performance is measured directly in years[6]. Several regularisation techniques are used to improve generalisation. Dropout (0.3) helped reduce overfitting, and EarlyStopping (patience 10) further prevented it by restoring the best validation weights. ReduceLROnPlateau lowers the learning rate by a factor of 0.5 when validation loss stagnates (patience 5, minimum learning rate 1×10^{-7}), enabling finer optimisation in later stages. [7]

Performance Analysis :



Interpretation of Results

Gender (classification)

- Training loss decreases from about 0.7 to ~0.17, and accuracy rises from ~0.55 to ~0.95.
- Validation loss drops to ~0.26–0.28 and accuracy to ~0.89–0.90, then plateaus.
- The small, stable train–val gap (a few percentage points in accuracy) shows the model fits the data well, with only mild lateepoch overfitting and good generalisation.

Age (regression)

- Training MAE decreases from ~17 to ~4.8 years, while validation MAE drops from ~17 to around 7.0–7.2 years before flattening.
- The validation curve closely tracks the training curve, with a consistent gap of about 2–2.5 years.
- This indicates the model has enough capacity (no underfitting), learns meaningful age features, and only slightly overfits the training data while still generalising reasonably well.

4. Model B — Transfer Learning Model

Model B: Transfer Learning with MobileNetV2

Architecture and MultiOutput Design

Model B is a multitask CNN built on top of a pretrained MobileNetV2 backbone for joint gender classification and age estimation. I used the Keras implementation of MobileNetV2, initialised with ImageNet weights and with the original classification head removed (include_top=False, input size 128×128×3). The convolutional backbone has been initially frozen (trainable = False), so it acts as a fixed feature extractor for the UTKFace images.[8]

The feature maps from MobileNetV2 are compressed using a Global Average Pooling (GAP) layer to obtain a single feature vector per image. On top of this, I added a shared head consisting of a Dense layer with 256 ReLU units and dropout (rate 0.5) before and after the dense layer. This head is shared by both tasks and regularised to reduce overfitting on the relatively small training set (~5k images).[9]

Compared to Model A, Model B includes a small gender-specific Dense layer after the shared head. This extra layer allows the network to learn features that are tailored to the gender task (for example, hairline or facial-hair cues) without forcing the age branch to use the same representation, which empirically helped gender accuracy while keeping age MAE stable.

GAP -> (1280) -> Dropout -> Dense(256) -> Dropout -> [Gender: Dense(1,sigmoid) | Age: Dense(64) -> Dense(1,relu)]

Choice of Backbone and Transfer Learning Rationale

MobileNetV2 is chosen as the backbone instead of heavier architectures such as VGG16 or ResNet for three main reasons:

Parameter efficiency and overfitting control

MobileNetV2 is a lightweight CNN that balances accuracy and efficiency, making it well-suited to our small (~5k images) UTKFace subset by reducing overfitting and keeping training and inference times manageable.

Comparison with VGG16 and ResNet

Compared to heavier backbones like VGG16 (~138M parameters) and ResNet, MobileNetV2 (~3.5M parameters) offers a much better trade-off between accuracy, model size, and speed, reducing overfitting risk and training cost for this small UTKFace-based age–gender multitask problem. [10]

Transfer learning from ImageNet

Starting from ImageNetpretrained weights allows the model to reuse generic low and midlevel visual features (edges, textures, shapes) that transfer well to facial images. The loss weights are set to 0.6 for gender and 0.4 for age so that the optimiser gives slightly more emphasis to improving gender accuracy while still controlling age MAE.[4]

To control overfitting and improve generalisation, I used:

1. Dropout (0.5) in the shared head to regularise the representation.
2. EarlyStopping on validation loss with restore_best_weights=True, so training stops when validation performance no longer improves and the best checkpoint is restored.
3. ReduceLROnPlateau to automatically lower the learning rate when validation loss plateaus, enabling finer updates in later epochs.
4. A staged training schedule (warmup + finetuning), avoiding large updates to the pretrained backbone at the start of training.

Training Strategy: Freezing, FineTuning, and TwoStage Schedule

Model B is trained in two stages, following standard transfer learning practice.

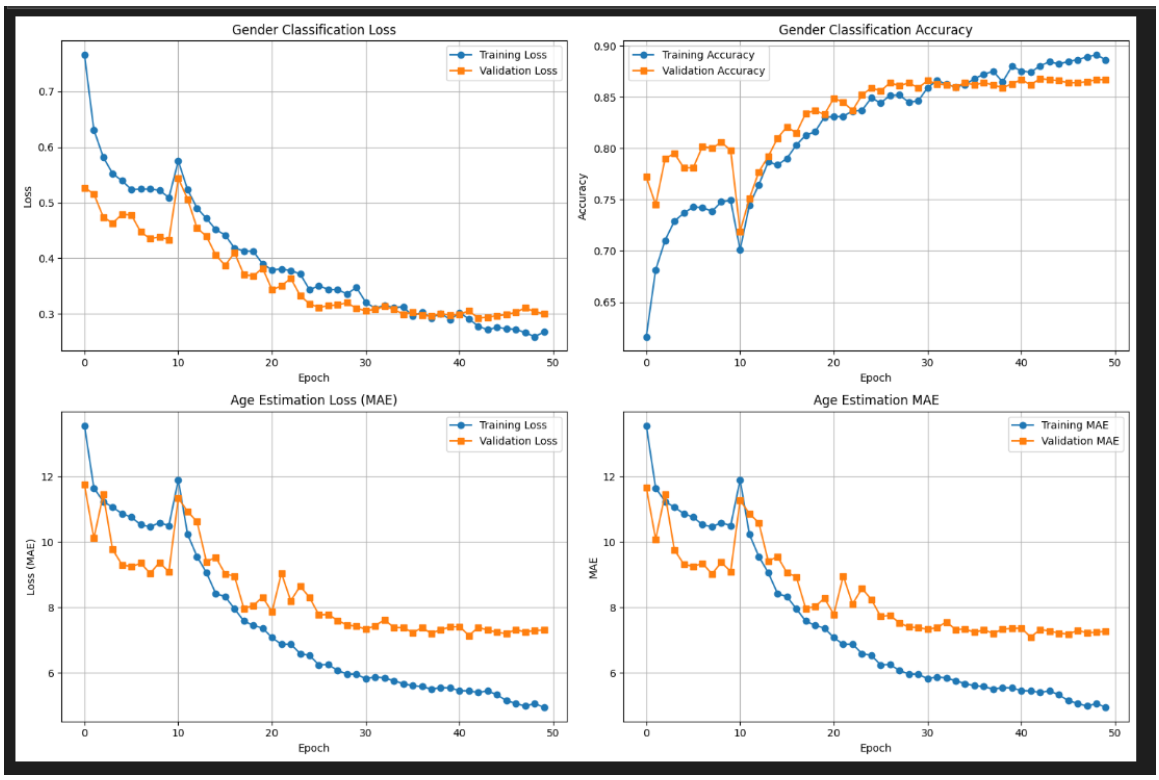
Stage 1: Head warmup (backbone frozen)

- For the first 10 epochs, I froze the entire MobileNetV2 backbone and trained only the new head using Adam (lr = 0.001) with the multitask loss (BCE for gender, MAE for age, weights 0.6/0.4), so that the head could first learn to use the fixed ImageNet features without disturbing the pretrained backbone.

Stage 2: Partial finetuning of the backbone

- After warmup, I partially unfroze the backbone by making approximately the last 30 layers of MobileNetV2 trainable, while keeping earlier layers frozen. These upper layers encode higherlevel, taskspecific features such as facial structure and texture, which benefit most from adaptation to UTKFace.
- The model is recompiled with a lower learning rate (0.0001) and training continues up to epoch 50 (or until early stopping). Using a lower learning rate in stage 2 prevents large updates that could destroy useful pretrained features, while still allowing gradual adaptation to the new dataset.

Performance Analysis :



Gender classification:

- Training accuracy increases from ~0.57 to ~0.88–0.90.
- Validation accuracy increases from ~0.77 to ~0.86–0.87.
- Training loss decreases from ~0.86 to ~0.27, while validation loss decreases to ~0.30.
- The train–validation gap is small and stable, indicating good generalisation with only mild overfitting.

Age estimation:

- Training MAE decreases from ~15.7 years to ~4.8–5.0 years.
- Validation MAE decreases from ~11.7 years to ~7.1–7.3 years and then plateaus.
- The gap between training and validation MAE stabilises around 2–3 years, which suggests that the model captures meaningful agerelated features without severe overfitting.

5. Summary, Comparison & Discussion

Aspect	Custom CNN (graph1)	MobileNetV2 (graph2)	Interpretation
Gender – training loss	Decreases from about 0.7 to ~0.17.	Decreases from ~0.86 to ~0.27.	Both models optimise well; Model A reaches a lower final training loss.
Gender – validation loss	Drops to ~0.26–0.28 and then plateaus.	Decreases to ~0.30.	Validation loss is slightly lower for Model A, indicating marginally better fit on unseen data.
Gender – training accuracy	Rises from ~0.55 to ~0.95.	Increases from ~0.57 to ~0.88–0.90.	Model A achieves higher final training accuracy, suggesting higher capacity.
Gender – validation accuracy	Increases to ~0.89–0.90 and plateaus with small train–val gap (few percentage points).	Increases from ~0.77 to ~0.86–0.87 with similarly small, stable gap.	Both generalise well, but Model A attains slightly higher peak validation accuracy.
Gender – overfitting behaviour	Small, stable gap; only mild late-epoch overfitting.	Small, stable gap; mild overfitting.	Both models are well-regularised; neither shows severe overfitting on gender.
Age – training MAE	Decreases from ~17 to ~4.8 years.	Decreases from ~15.7 to ~4.8–5.0 years.	Final training MAE is very similar; Model B starts lower but both end around 5 years.
Age – validation MAE	Drops from ~17 to ~7.0–7.2 years before flattening.	Drops from ~11.7 to ~7.1–7.3 years and then plateaus.	Both reach comparable validation MAE around 7–7.3 years.

The UTKFace tests were carried out on a smaller set of 5,000 images, which is only about 21% of the full aligned dataset. The MobileNetV2 model achieved about 89–90% test accuracy for gender classification and an age MAE of about 7 years on this subset. This shows that it can learn useful features even with little data. But training on fewer samples makes it harder for the model to cover rare age groups and demographic differences, i.e to generalize. The light augmentation used also means that the MAE number is low and could be better with stronger augmentation or training on the whole dataset.

Another problem is that the dataset is not distributed fairly distributed across ages, ethnicities, and genders, which could lead to bias and inconsistent performance across different demographic groups. In the real world, deployment would also have to deal with domain shift caused by changes in lighting, camera quality, and pose that aren't fully represented in the training data. These factors demonstrate how we need more diverse datasets, a way to judge fairness, and stronger training strategies for real-world use.

References

[1] Shorten, C. and Khoshgoftaar, T. (2019) 'A Survey on Image Data Augmentation for Deep Learning', Journal of Big Data, 6(60).

[2]TensorFlow (2024). tf.data: Build TensorFlow Input Pipelines. Available at: https://www.tensorflow.org/guide/data

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[5]Keras (2024) Keras API Reference. Available at: https://keras.io/api/

[6]Hyndman, R.J. and Koehler, A.B. (2006) ‘Another Look at Measures of Forecast Accuracy’, International Journal of Forecasting, 22(4), pp. 679–688.

[7]TensorFlow (2024) TensorFlow Callbacks: EarlyStopping and ReduceLROnPlateau. Available at: https://www.tensorflow.org/api_docs/python/tf/keras/callbacks.

[8]Pan, S.J. and Yang, Q. (2010)

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Deep Learning. MIT Press.

[10] Simonyan, K. and Zisserman, A. (2014) *Very Deep Convolutional Networks for Large-Scale Image Recognition.* Available at: https://arxiv.org/abs/1409.1556